

Team Number - 9

Dhawal Pawanarkar - 2025204005

Devansh Singh - 2025204007

Swarang Joshi - 2022114010

Introduction -

Pregnancy, specially for first time mothers, is one of the most overwhelming experiences a person can go through, and many women still don't have access to reliable and easy to understand digital support throughout their pregnancy journey. This literature review looks at how AI enabled digital platforms can be designed to help with this, covering areas like trimester wise education, pregnancy tracking, personalised health guidance, and peer support communities, all under the subject of User Interaction and Usability of Digital Products. A total of eighteen papers were selected using keywords such as First-Time Mother Experience, Maternal Digital Health, AI-Enabled Prenatal Care, Pregnancy Tracking Systems, Digital Peer Support Communities, Usability and Accessibility in Health Applications, Visual Learning Tools, HCD for Maternal Care, and Personalised Health Advice.

Keywords Used for Paper Selection

first-time mother experience, maternal health education, prenatal care journey, pregnancy information visualization, pregnancy tracking and self-monitoring, digital peer support communities, usability and accessibility in health applications, human-ai interaction in healthcare, ai-driven personalised health guidance, maternal digital health, ai-enabled prenatal care, virtual peer support, digital literacy, visual learning tools, and hcd for maternal care

HMW

How might we design an AI-enabled intuitive digital platform, including digital support groups or communities for first-time mothers, that provides easy-to-understand educational tools to learn and visualise each trimester of pregnancy, personalised health advice, pregnancy tracking, understanding of prenatal care requirements, and enables mothers to share experiences and seek peer support?

Revolutionizing Maternal Health: The Role of Artificial Intelligence in Enhancing Care and Accessibility

Problem Addressed:

Delayed detection of pregnancy complications along with lack of access to quality care/support leads to high female mortality and morbidity. AI can bridge the gap in maternal healthcare.

Method/Approach:

Review of AI applications in maternal health include - predictive analytics, ML for risk mitigation, remote monitoring, telemedicine, and virtual assistants(chatbots).

Key findings/Contributions:

- a. AI ML helps in early detection of complications using predictive modelling.
- b. Remote monitoring and wearable devices enable continuous tracking and timely alerts
- c. VA can help in knowledge enhancements and scheduling of medicines/appointments/communications.

Limitations:

Data privacy concerns. Lack of Explainability/ XAI to understand the decisions. Human oversight (Human in the loop) required as the systems can't be fully autonomous yet.

A Chatbot System for Supporting Women and Families during Pregnancy

Problem Addressed:

Mothers + families lack personalised feedback and guidance which is often required for first pregnancy. Realtime / Accessible 24/7 framework isn't available throughout and existing systems are often overloaded

Method/Approach:

Design and Implement "PregoBot" - AI chatbot (conversational NLP) to provide tailored, personalised feedback, offer support and guidance, build community support for users during pregnancy.

Key findings/Contributions:

PregoBot offers personalized info based on user input

NLP applied conversational interaction, making it more user-friendly.

Continual Learning- The system continually learns from interactions, adapting to user needs.

Limitations:

Descriptive rather than empirical. No actual user feedback reported. No performance evaluation - either human / without human. Real world effectiveness is not validated.

Women's Preferences and Willingness to Pay for AI Chatbots in Women's Health

Problem Addressed:

Rapid Development of various AI enabled tech for health education but there is no proper concrete study on user needs. Very little is known with regards to understanding which features are most useful to women/ what they are willing to pay for.

Method/Approach:

Discrete Choice Experiment (DCE)- 957 women in China.

Six chatbot attributes tested: response accuracy, legibility, cost, background data collection, usefulness of information, and content focus. Preferences and willingness to pay (WTP) estimated using a conditional logit model.

Key findings/Contributions:

Most preferred attributes - High response accuracy, very easy-to-understand information, and practical utility

Participants preferred free chatbots over paid ones. (CN ¥0/month).

But users are willing to pay more if most preferred attributes are offered.

Limitations:

Single hospital study - lack of generalisation, pref may vary across demographics and regions. Trust/ perception of ai and health applications not explored.

Can AI Bridge or Widen Maternal Health Inequities?

Problem Addressed:

AI applications are being rapidly developed, its effect on reducing vs increasing existing disparities not studied.

Method/Approach:

Conceptual framework on how design, governance, and implementation of AI leads to equity outcomes in maternal health.

Key findings/Contributions:

AI has potential to reduce inequities if created with inclusive data practices and being welcoming to wider audience.

Lack of careful governance, AI could amplify disparities, particularly in Low Mid Income Countries.

Four domains shaping equity: inclusive data, equitable governance, participatory design, local capacity.

Limitations:

Not an empirical study. It outlines theory but no quantifying impact. Evidence of outcomes not established.

Practical Guidance for the Development of Rosie, a Maneuverable AI Health Education Chatbot

Problem Addressed:

Community of colour experiences higher mortality rates than average. Need for accessible, reliable, and culturally sensitive health information

Method/Approach:

Developed "Rosie," a health-ed chatbot, using approach that combined tech development, ML, and community outreach. live demo and pilot testing with 109 pregnant women and new mothers, primarily from colour communities.

Key findings/Contributions:

high demand for the tool, with 75% of users searching health info weekly and 90%+ expressing likelihood to use the app. most common questions related to infant development, nutrition, and symptom management. developing health tech "with" the community rather than just "for"

Limitations:

relatively small sample size (109) and was geographically limited US - (DC/Maryland/Virginia area)

Pregnancy and Health in the Age of the Internet: A Content Analysis of Online "Birth Club" Forums

Problem Addressed:

Most research on how pregnant women use internet rely on self-reported data. may not accurately reflect actual info searching behavior. Aim to understand what women discuss and seek from online peers.

Method/Approach:

content analysis using NLP (Latent Dirichlet Allocation) on 262,238 posts from "birth club" forums on WhatToExpect.com - period of 1 year.

Key findings/Contributions:

Contrary to the assumption- forums are mainly for emotional support. the largest topic category - maternal health (45%), next - baby-related topics (29%). Dominant topics shifted by trimester: Miscarriage (1st trimester), Labor/Delivery (3rd trimester), and Baby Sleep (Postpartum). findings suggest significant informational needs in clinical domains (like labor and miscarriage)

Limitations:

study only analyzed the initial posts of threads, not the full conversations. relied on data from a single website - not generalizable enough.

AlJaberi, H. (2018). Developing Culturally Sensitive mHealth Apps for Caribbean Immigrant Women to Use During Pregnancy: Focus Group Study. *JMIR Human Factors*, 5(4), e29.

Problem Addressed:

The existing mobile health app usually fail to meet the need of the minority population (here we talked about Caribbean immigrant women in the USA) they face many challenges and are often ignored by the general mobile health apps.

Method / Approach:

The researchers did 3 focus groups which had 12 low income Caribbean immigrant women living in south Florida. They used the concept of “pregnancy ecology”. They explored women’s specific need, stressors and technology habits.

Key findings / Contributions:

they felt alienated by the mainstream apps that was made for wealthy white people. The app lacked their cultural diet. Family separation was a main stressor. The study contributed a framework for recommending the mobile health app to address mothers emotional needs, socioeconomic realities, clear up cultural misconceptions.

Limitations:

the sample size was small (just 12 participants). They excluded non english speaking families which failed to capture language barriers that affect technology adoption

Rivera Rivera, J. N., Smith, M., Mathur, S., AuBuchon, K. E., Thomas, A. D., & Arem, H. (2024). Development and refinement of a chatbot for birthing individuals and newborn caregivers: Mixed methods study. *JMIR Formative Research*, 8, e54315.

Problem addressed:

the fourth trimester (the last 42 days following childbirth) carry high risk for the mother and the infant. There is a gap in providing continuous postpartum education and connection to care for the warning sign after hospital discharge.

Method / Approach:

it was a mixed method study that evaluated two chatbots (postpartum and newborn) which was deployed to over 4000 patients. They analyzed engagement data and conducted surveys and few semi structured interviews. The group was made of english and spanish speaking users

Key Finding / Contributions:

strong engagement with the chatbot with over 60% of the patients opening the messages. Participants recommended to extend the intervention beyond 6 weeks and adding deeper content on mental health.

Limitations:

relied heavily on patients recall which risked confusion between chatbot and other health app that they used.

Shrivastava, R., Singhal, M., Gupta, M., & Joshi, A. (2023). Development of an artificial intelligence–guided citizen-centric predictive model for the uptake of maternal health services among pregnant women living in urban slum settings in India: Protocol for a cross-sectional study with a mixed methods design. *JMIR Research Protocols*, 12, e35452.

Problem addressed:

Most of the Pregnant women living in urban slums are a high risk group. The barriers lead to underutilization of maternal health services. Which result in preventable maternal and child mortality and low birth weight

Method / approach:

they enrolled 225 pregnant women in south delhi slums. Using android based tools, they collected data of their sociodemographics, diet and clinical history.

Key findings / Contributions:

contributed a functional AI guided, human centric support system that predicts complications, delivery methods. The ai enabled policymakers to prioritize resource allocation and improve service for the the marginalized communities

Limitations:

enrollment challenges and the potential loss of follow up because of covid 19 pandemic. The small sample size restricted to a single geographic limits the generalization of the AI model.

Bailey, E., Nightingale, S., Thomas, N., Coleby, D., Deave, T., Goodenough, T., Ginja, S., Lingam, R., Kendall, S., Day, C., & Coad, J. (2022). First-time mothers' understanding and use of a pregnancy and parenting mobile app (The Baby Buddy App): Qualitative study using appreciative inquiry. *JMIR mHealth and uHealth*, 10(11), e32757.

Problem addressed:

the quality of mHealth app still varies globally. The limited evidence of how well they affect the maternal well being is also a problem.

Method / approach:

researchers conducted semi structured telephone interviews and a focus group with 17 first time mothers in UK. they identified what worked well and what could be improved regarding their use of the NHS approved baby buddy app

Key findings / Contributions:

mothers said that the app is reliable. The information is easy to access. They trusted the content because it was backed by health care professionals. However mothers still preferred face to face reassurance for the complex issues.

Limitations:

the sample size was very low merely accounting for just 17 women who were well educated and white.

Jiang, L., & Zhu, Z. (2022). Maternal mental health and social support from online communities during pregnancy. *Health & Social Care in the Community*, 30(6), e6332–e6344.

Problem addressed:

lack of prenatal support is a significant risk factor. While traditional interventions exist, it is unclear how much digital information in online maternity groups is generally perceived as social support and whether it correlates with the health and birth outcome

Method / approach:

researchers surveyed 500 users who all were in a large Chinese online maternity community. They calculated perceived social support (PSS) and evaluated mental health using Edinburgh Depression Scale.

Key findings / Contributions:

the online information in the groups was perceived as genuine social support. Higher PSS score were associated with lower levels of depression. Greater online support correlated with heavier newborn birth weight.

Limitations:

the study lacked the direct comparison with offline, face to face evaluation in the support efficacy.

Small, Z., Thompson, S. E., Sharma, A., Majumdar, S., Thout, S. R., Praveen, D., & Hirst, J. E. (2023). Mobile health (mHealth) interventions for health promotion during the perinatal period in India: a scoping review. *BMJ Global Health*, 8(9), e012351.

Problem addressed:

the prenatal, maternal mortality rate are very high in india. There are multiple mHealth app deployed which try to improve the outcome but lack clarity on how they can best support women in india's diverse culture.

Method / approach:

researchers screened 1783 records across 4 databases. Identified 29 studies and 22 mobile health app interventions which target pregnant / postpartum women in india

Key findings / Contributions:

the interventions relied on one way communication. Shared phone ownership (often controlled by husband), low literacy and heavy household responsibility were issue.

Limitations:

the review only included english language. They didnt assess the health economic impacts or cost benefit of the mhealth interventions.

Müller, A. et al. (2023). Usability and perception of a wearable-integrated digital maternity record smartphone app for pregnant women

Problem addressed:

The paper highlighted the experience of women based on how they interacted with various digital maternity record applications, it also addressed how they become more complex when some if not all of these applications also had a feature to integrate some of the functions which are being offered by various wearable devices like heart rate monitoring and also monitoring of pregnancy.

Method/approach:

The authors formulated a guided prototype app for pregnancy which then tried to integrate these wearable device's data and attempted to conduct a study called "think-aloud" user tests and some questionnaires which are also called SUS, UEQ with the participants involved.

Key findings/contributions:

Based on the method mentioned above, the application did achieve a mean score around 79, which concluded to 79 thus indicating a high usability and then it was rate somewhat above average based on the following parameters -

1. Attractiveness
2. Efficiency
3. Clarity

But the application performed poorly on 4. Novelty metric which was then termed as below average. The study pointed that a design based on user centred, guided medically can achieve a highly effective / usable pregnancy application.

Limitations (noted by authors / implied):

The application was only evaluated based on prototype, further it had a limited sample size, also it focused more on one health system which in turn reduced generalisability.

Chung, S. et al. (2025). Chatbot to support the mental health needs of pregnant and postpartum individuals: Design, development, and usability evaluation of “Moment for Parents”. JMIR Mental Health.

Problem addressed:

The paper pointed on how the maternal mental health disorders are very common among pregnant and other postpartum women but some of the issues like stigma and limited access prevent many from getting the support they need.

Method/approach:

The authors used a human centred design again starting with phase one wherein they conducted some ethnographic interviews along with pregnant and postpartum users to understand their core needs and other preferences which was then followed by a phase two which involved continuous pilot testing of the chatbot prototype with some of real users to check the usability engagement and relevance of the prototype.

Key findings/contributions:

The individual interviews helped them shape content directly. Pilot testing concluded that a curated, conversational chatbot interface can deliver somewhat acceptable and usable mental health education and provide support, proving the feasibility of AI style chat tools in perinatal care.

Limitations (noted by authors / implied):

Since it was an early stage pilot evaluation the results depended on some of the self selected participants and the chatbot too focused mainly on mental health rather than the full prenatal care hence limiting generalization only to physical health aspects.

Kovács, B. et al. (2025). User experience with pregnancy tracker mobile apps: A comment-based qualitative study.

Problem addressed:

The paper points to the fact that around millions of women use many popular pregnancy tracking apps daily, but there is limited understanding of their real world user experiences especially from looking at the appstore comments and reading on online communities.

Method/approach:

Performed a qualitative analysis of user comments on popular pregnancy apps to explore some behaviors, expectations, pain points and the role of social and/or emotional factors in shaping UX.

Key findings/contributions:

Positive experiences came from app usefulness, high quality information, advanced features, privacy protection, and solving practical problems for users.

Negative experiences grouped around poor visualization, technical glitches, repetitive/irrelevant content, scary peer stories, gendered assumptions, and misinformation.

Limitations (noted by authors / implied):

The comment-based data likely over-represents very satisfied or dissatisfied users, and the study could not link findings to actual health outcomes or long-term app adherence.

McCarthy, M. et al. (2022). First-time mothers' understanding and use of a pregnancy and postnatal smartphone app: Baby Buddy. *Midwifery*.

Problem addressed:

The paper highlights the fact that even though there are a large number of pregnancy and postnatal apps available to women today there was very little understanding of how first time to be mothers specifically interact with these apps what drives them to use them and what real benefits they gain from using them. Mothers are in a particularly unique position because they have no previous experience of pregnancy or childbirth to rely on or look upto which makes them much more dependent on outside sources of information from the mothers who have been through the experience before.

Method/approach:

The researchers carried out semi structured telephone interviews and focus groups with first time mothers who had used the Baby Buddy app either during their pregnancy or after giving birth. This info gathered from these sessions was then analyzed using a method called “applied thematic analysis” which is a flexible qualitative approach that allows key themes to come out from the data while still staying closely connected to what the participants actually said and experienced this method hence allowed the researchers to capture detailed usage patterns, emotional reactions.

Key findings/contributions:

The app was effective at helping the first time mothers feel more confident by giving them reliable daily info via pop-ups and small, easy to relate content that was matched to their current stage of pregnancy or postnatal period, also when it came to more complex emotionally and sensitive medically specific concerns , the users preferred to speak directly with a healthcare professional in person which suggested that the app works best as a supporting tool.

Limitations (noted by authors / implied):

Conducted entirely within UK setting based on the National Health Service context, this implied that the findings may not be directly applicable to the broad category of mothers living in other countries where different healthcare systems exists and different cultural backgrounds or access to services are not relatable.

Rosario, A. et al. (2021). Establishing content for a digital educational support group for new adolescent mothers in the Dominican Republic: a user-centered design approach. *BMC Pregnancy and Childbirth*.

Problem addressed:

The paper highlights the fact that adolescent mothers particularly those living in resource limited settings, face very specific challenges after childbirth such as social isolation, limited knowledge about newborn care, breastfeeding, and managing their emotional wellbeing. The paper addresses the question of what content should actually be included in a digital educational support group designed for this group, since the informational needs of adolescent mothers are quite different from those of older mothers.

Method/approach:

The researchers followed a two phase user centered design approach. In the first phase conducting group sessions and personal interviews with adolescent mothers to understand their experiences till now and their everyday challenges and which topics they considered overall important. In the second phase, participants took part in an activity where they ranked each topic by their relevance and how important it was helping the team identify what content mattered.

Key findings/contributions:

The study concluded that some of the topics which were most important to an adolescent mothers were breastfeeding and managing emotions and newborn care and nutrition. These ranking of topics directly shaped the content structure of the proposed digital support group. The study also validated that WhatsApp is a practical and feasible platform for reaching adolescent mothers in the country in which they live.

Limitations (noted by authors / implied):

The findings are very specific to adolescent mothers in the Dominican Republic which is hence shaped by the country's particular cultural norms and healthcare infrastructure, and socioeconomic conditions. This limits how directly the results can be applied to other countries or different age groups. The study also noted that smartphone and mobile internet access was not at all consistent among participants, which presented a real problem to scale this kind of digital intervention more broadly.

(2025). Designing, validation and evaluation of an mHealth app for prenatal health promotion using the ADDIE model. *Journal of Medical Internet Research*.

Problem addressed:

The paper pointed to the fact that there is a clear gap in understanding what users actually expect from health applications that are built for prenatal health education and behaviour change. Many of these are built without any proper design process which accounts for user needs, cultural factors and learning goals as a result of this they most likely fail to bring about any real behaviour change or keep the users constantly engaged.

Method/approach:

The authors made use of an instructional design framework called the “ADDIE model”, which is - Analysis Design Development Implementation Evaluation and used it as the main part for their development process. In the analysis phase - interviews were carried out with pregnant women alongside a literature review to understand the user needs and barriers. Similarly in the design phase an expert focus group from fields like 1.Obstetrics 2.Midwifery 3.Health Education was formed to validate the content and interface decisions along with cultural relevance of the app.

Key findings/contributions:

The study found that the key UX and cultural features were needed for a prenatal app to actually drive the behaviour change which included personalized goal-setting, culturally sensitive language, visual learning tools, simple navigation and voice guidance and offline access for users with limited internet. It also showed that skipping a proper analysis phase during development is the reasons why apps end up feeling generic and irrelevant to users.

Limitations (noted by authors / implied):

The finding from papers only covers the Analysis and Design phases of the ADDIE this means that the app was not yet tested or deployed to real users at the time of publishing and no actual individual interaction info was available.

Paper	Core Focus	Trend It Reflects	Similarity With Other Papers	Key Difference From Others	Research Gap It Highlights
Mapari & Shrivastava (2024)	AI in maternal health overview	AI increasingly used for maternal health support	Shares with 2, 3, 5, 8, 14 — all explore AI-driven tools for maternal care	Entirely theoretical; no users involved or empirical data collected	Lack of real-world validation and XAI explainability in AI maternal tools
Bhaskar & Kunchala (2023)	PregoBot — pregnancy chatbot	AI/chatbot tools for personalised pregnancy support	Shares with 1, 5, 14 — chatbot design for pregnant women	System design paper with no empirical user testing or feedback reported	No platform integrates chatbot, community, and tracking together
Wang & Min (2024)	Women's preferences and WTP for AI chatbots	User preferences shaping AI health tool design	Shares with 10, 15, 16 — accuracy and clarity are top user priorities	Largest quantitative sample (957); uses DCE which no other paper uses	Trust and cultural perception of AI in health largely unexplored

Laguitan & Bernardino (2024)	AI equity in maternal health	Recognition that AI can widen or bridge disparities	Shares with 7, 9, 12 — all highlight underserved populations being poorly served	Purely conceptual framework; no empirical measurement of equity outcomes	AI equity impact in maternal health is theorised but never empirically measured
Mane & Doig (2023)	Rosie — AI chatbot for communities of colour	Community co-design and cultural sensitivity	Shares with 7, 17, 18 — participatory, user-centred design approach	Only paper that explicitly co-designs with communities of colour	Culturally sensitive AI tools for diverse maternal populations understudied
Wexler & Davoudi	Online birth club forums	Online communities used for health info, not just emotional support	Shares with 11, 17 — peer communities hold significant maternal health value	Only paper using NLP/LDA on 262,238 forum posts at scale	Trimester-specific information needs within communities are rarely mapped
AlJaberi (2018)	Culturally sensitive mHealth for immigrant women	Marginalised groups underserved by mainstream apps	Shares with 4, 9, 12 — equity and cultural sensitivity in mHealth	Focuses on immigrant women specifically; uses "pregnancy	Cultural and language diversity rarely built into maternal app design

				ecology" framework	
Rivera Rivera et al. (2024)	Postpartum chatbot for birthing individuals	AI tools for continuous postpartum care and support	Shares with 2, 14 — chatbot-based perinatal support	Largest chatbot deployment (4000+ patients); focuses on fourth trimester specifically	Most AI tools cover only one stage; full pregnancy-to-postpartum journey not covered
Shrivastava et al. (2023)	AI predictive model for maternal health (India)	AI for underserved and marginalised communities	Shares with 4, 12 — mHealth in India and low-resource settings	Only paper building an AI predictive model specifically for urban slum populations in India	Indian urban first-time mother context for AI platforms is unexplored
Bailey et al. (2022)	Baby Buddy app for first-time mothers	Trust and reliability drive app engagement	Shares with 3, 15, 16 — users prioritise accurate, trustworthy content	Focuses on NHS-approved app; only paper using appreciative inquiry method	Long-term engagement data post-app use is entirely missing

Jiang & Zhu (2022)	Online maternity community and mental health	Peer support communities improve measurable health outcomes	Shares with 6, 17 — digital peer communities hold real maternal health value	Only paper directly correlating online peer support with birth weight outcomes	Peer support and AI personalisation have never been studied together
Small et al. (2023)	mHealth during perinatal period in India	One-way communication dominant in Indian mHealth	Shares with 4, 7, 9 — marginalised groups face structural barriers to digital health	Scoping review of 29 studies specific to India; highlights shared phone and low literacy barriers	India-specific mHealth gaps for first-time mothers are largely unaddressed
Müller et al. (2023)	Wearable-integrated maternity app usability	User-centred, medically guided design improves usability	Shares with 15, 16, 18 — usability and UX evaluation of pregnancy apps	Only paper integrating wearable device data with a maternity record app and evaluating via SUS/UEQ	Wearable-integrated tracking with peer and community features not yet studied together

Chung et al. (2025)	Perinatal mental health chatbot	HCD shapes effective AI chatbot design	Shares with 2, 5, 8 — chatbot for perinatal support; shares with 17, 18 — HCD approach	Only chatbot paper using full ethnographic HCD with two distinct design phases	Mental health chatbots do not cover full prenatal care spectrum
Kovács et al. (2025)	UX of pregnancy tracker apps via comments	Real-world UX pain points in pregnancy tracking apps	Shares with 3, 10, 16 — trust, accuracy, and visualisation critical to positive UX	Only paper using app store comment analysis (NLP) as the primary data source	Poor visualisation in pregnancy apps widely noted but rarely studied as a design intervention
McCarthy et al. (2022)	Baby Buddy — first-time mother engagement	First-time mothers need confidence-building, not just information	Shares with 10 — both study Baby Buddy; shares with 3, 15 — trust and reliability key	Only paper using applied thematic analysis on first-time mother focus groups for postnatal app use	No study tracks first-time mother engagement longitudinally across all trimesters

Rosario et al. (2021)	Digital support group content for adolescent mothers	Participatory design for digital maternal communities	Shares with 5, 7, 18 — user-centred design involving target women	Only paper focused on adolescent mothers in a low-resource Caribbean setting; validates WhatsApp as platform	Effectiveness and usability of digital peer groups after design phase never evaluated
Bayrami et al. (2025)	Prenatal mHealth app via ADDIE model	Structured instructional design improves relevance and personalisation	Shares with 13, 15, 16 — UX and usability evaluation; shares with 5, 14, 17 — HCD approach	Only paper applying full ADDIE instructional design model to prenatal health app development	Design-stage only; no post-deployment user testing; Iran-specific cultural tailoring limits generalisability

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